Real-time Emotion Recognition and Expression for Avatar-Mediated Mental Health Counseling: Improving Nonverbal Communication and Emotional Response

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***Abstract*—Effective nonverbal communication is crucial in mental health counseling, and avatar-mediated communication offers potential solutions to enhance the nonverbal communication aspect. In this paper, we propose a real-time emotion recognition and expression model for avatar-mediated counseling and analyze its impact on the emotional experience and rapport building of clients. We recruited 60 college students, assigned 30 to a control group and 30 to an experimental group. The experimental group participated in avatar-mediated counseling sessions where the avatar responded with facial expressions that matched the emotions conveyed by the participants, while the control group received counseling sessions from an avatar without facial expressions. Our preliminary results showed that the experimental group perceived significantly higher sense of presence, immediacy, interactivity, and similarity with the avatar compared to the control group. Additionally, the experimental group reported significantly higher levels of satisfaction, trust, and rapport compared to the control group. Our study demonstrates the potential benefits of using avatars with real-time emotion recognition and expression in mental health counseling.**

Keywords— nonverbal communication, avatar-mediated counseling, emotion recognition and expression model, mental health

I. Introduction

Mental health is a critical aspect of overall well-being, and effective communication between a counselor and a client is essential for mental health treatment. Nonverbal communication, such as facial expressions, body language, and tone of voice, plays a crucial role in conveying emotions and building rapport between the counselor and the client [1]. However, some clients may find it challenging to express their emotions in traditional counseling settings, making it difficult for counselors to understand their needs fully. In recent years, artificial intelligence (AI) media agents have emerged as a potential solution to this issue, as they can provide cognitive and emotional empathy to individuals experiencing mental health issues, potentially improving mental health outcomes. By combining sensory synchronicity, AI media agents can offer users a sense of presence, reality, immediacy, and interactivity, making them suitable for a range of mental health scenarios and requests. While previous studies have examined the effects of AI media agents on mental health, most have focused on the agents' physical appearance and technical and spatial design features, neglecting their verbal and nonverbal communication characteristics and their impact on human cognition, motivation, and emotion [2]. Nonverbal communication is especially crucial in mental health counseling, as it can convey emotions, attitudes, and intentions that may be difficult to express through words. Facial expressions, tone of voice, and body language are all important components of nonverbal communication that can affect the therapeutic relationship and the client's overall experience [3]. Moreover, individuals with mental health issues may struggle with verbal communication or have difficulty expressing their emotions, which underscores the need for effective nonverbal communication in counseling.

In this regard, media agents with real-time emotion recognition and expression capabilities offer a promising avenue for enhancing nonverbal communication in mental health counseling. By providing users with a sense of presence, reality, immediacy, and interactivity, media agents can create an immersive and engaging environment that enables effective communication and rapport building between counselors and clients. Furthermore, media agents can respond to users' emotional cues and provide empathetic feedback in real-time, which can enhance the therapeutic alliance and improve mental health outcomes. Therefore, investigating the impact of media agents' nonverbal communication characteristics on mental health counseling is an important and timely research topic.

II. Literature Review

A. The Impact of Facial Expressions on Counseling

In [4], it was established that counselor facial expressions play a crucial role in building rapport with clients. However, the use of human counselors can have its limitations, as individual differences in facial expressions and other personal characteristics can influence the counseling process. Additionally, privacy concerns may prevent some individuals

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from fully expressing their emotions during in-person counseling sessions. This is where the use of avatars for counseling can be beneficial, as they offer a safe and private space for clients to express their emotions.

Moreover, avatars can provide a more standardized and controlled counseling experience, as the avatar's facial expressions can be programmed to respond in specific ways to client emotions. This is in contrast to human counselors, where facial expressions may be more difficult to control or regulate. Therefore, exploring the potential of avatar-mediated counseling as a viable alternative to traditional counseling is important.

However, it is essential to determine if the same psychological mechanisms that apply to in-person counseling also apply to avatar-mediated counseling. Specifically, it is important to investigate whether the facial expressions of an avatar can elicit similar emotional responses and build rapport in the same way that a human counselor can. Our study aims to emphasize the need for further research to explore the potential of avatar-mediated counseling and its impact on client satisfaction, rapport, and trust.

B. *Avatar-mediated Emotion Conveyance*

Prior research has shown that computer-generated avatars are effective at conveying emotions and can impact human emotions in various ways. Reference [5] conducted research on the use of performance-driven facial animation to convey emotional states in facial avatars. They found that computer-generated avatars can effectively convey emotions and affect human emotions in different ways, and that nonverbal communication through avatars was effective in conveying emotions and providing a sense of presence, realism, and interactivity in the communication process.

Reference [6] conducted a study on the affective valence and arousal of avatar facial expressions. Their findings showed that avatar facial expressions had a significant effect on the emotional responses of participants, particularly with respect to affective valence and arousal. However, one of the limitations of their study was that the participants were not engaged in a counseling context.

Overall, both studies provide important insights into the potential of avatars for conveying emotions and affective responses. However, they also have their own limitations and do not directly address the use of avatars for mental health counseling. This emphasizes the necessity for additional investigation in this field, particularly regarding the implementation of avatars for nonverbal communication in mental health counseling.

III. METHOD

A. *Avatar Design and Environment:*

We designed a 3D computer-generated avatar with customizable facial expressions and a range of emotions, including neutral, happiness, sadness, and anger. The avatar was designed to look human-like, with neutral facial features and a consistent outfit, and was created to resemble a mental health counseling office environment. In order to simulate a more realistic counseling session and enhance the user's sense of empathy and connection with the avatar, the avatar's responses were designed to reflect four emotions: neutral, sad, angry, fear and happy.

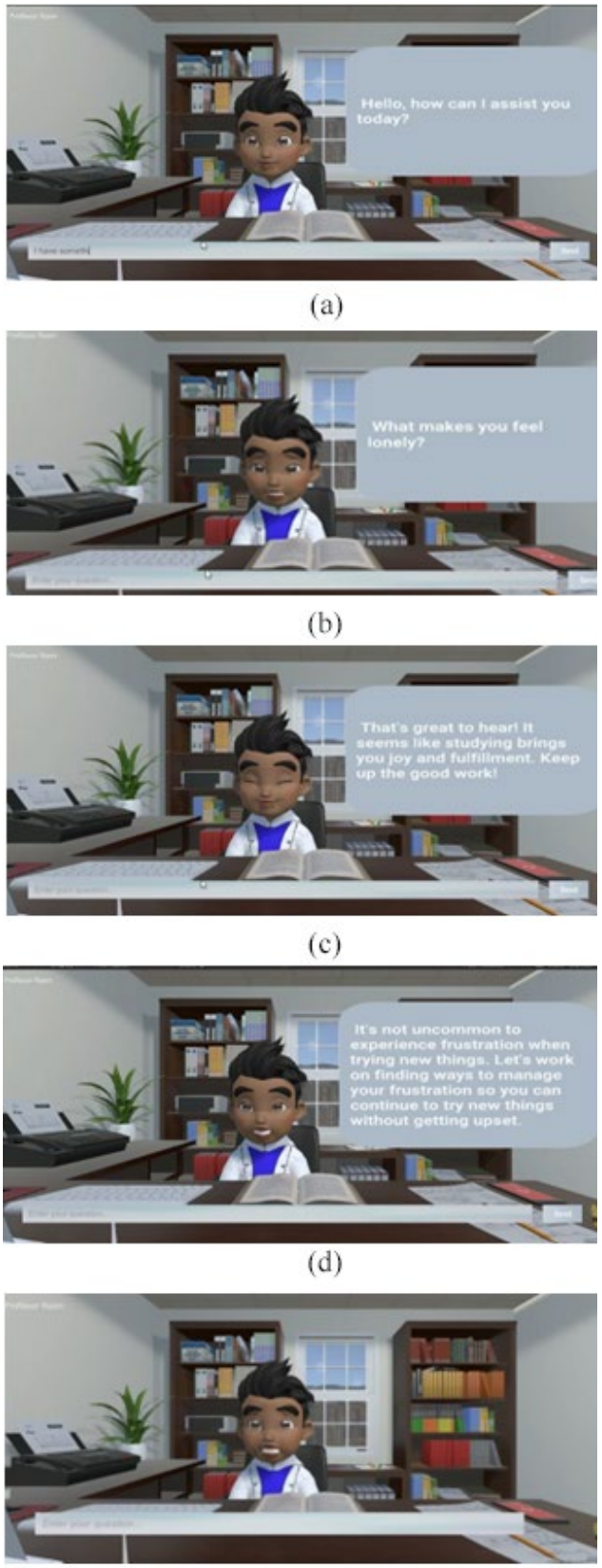

Figure 1. (a) Neutral response (b) Sad response (c) Happy response (d) Angry reponse (e) Fear reponse was expressed on the face of the avatar, with the facial expression changing instantly in response to comments submitted by the user via chat.

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When a user greeted the avatar with a common greeting such as "hello," the avatar would respond with a neutral expression. However, if the user expressed a negative emotion such as feeling lonely or frustrated, the avatar would respond with a sad or angry expression, respectively. On the other hand, if the user expressed a positive emotion such as making progress or feeling optimistic, the avatar would respond with a happy expression.

To enable the avatar to recognize and respond to emotions, we fine-tuned the chatGPT language model using 'patient emotion data.' This dataset was derived from real clinical psychotherapy sessions, in which the patient's spoken language was matched to their emotional state.

For the fine-tuning process, we first transformed this dataset into a machine-readable format, specifically a JSON format, where each instance consists of a 'prompt' and a 'completion.' The 'prompt' is the patient's utterance, and the 'completion' is the therapist's empathetic response marked with the corresponding emotion. For example, if a patient said, 'The weather is so clear and transparent... it feels like fragile glass that's about to break,' the statement would be paired with an empathetic response marked with 'sadness', like this:

{ "prompt": "The weather is so clear and transparent... it feels like fragile glass that's about to break",

"completion": "3 I see that you're feeling quite fragile, almost as if you might break like the glass you described. Would you like to talk more about what's causing you to feel this way?" }

In this instance, the number '3' in the completion represents the avatar's emotional response as sadness. We used hundreds of these prompt-completion pairs to fine-tune the chatGPT model. The model learns to generate appropriate emotional responses given the patient's utterance as a prompt.

The Emotion-Response module in our model then takes the 0~4 digits and utilizes this information to set proper facial expressions on the avatar. The construction of the avatar's facial expressions is based on the blend shape interpolation and Facial Action Coding System (FACS) described by Reference [7].

By fine-tuning the model and integrating emotional facial expressions in this way, we aimed to make the avatar more capable of empathizing with the emotions expressed by the user during the counseling session. Figure 1 illustrates the four possible responses of the avatar in relation to the user's input, including neutral, sad, angry, fear and happy responses.

B. Participant Recruitment

We recruited 60 undergraduate students to participate in the study, 30 of whom were assigned to the avatar-mediated counseling condition, while the other 30 were assigned to a control condition with a non-responsive avatar. Participants were screened for any history of mental health disorders and were required to provide informed consent before participating.

We recruited 60 undergraduate students to participate in the study, with 30 assigned to the experimental group receiving avatar-mediated counseling sessions with an avatar that provided real-time emotion recognition and expression, while the other 30 were assigned to a control group receiving counseling sessions from an avatar without emotional expression. Participants were screened for any history of mental health disorders and were required to provide informed consent before participating.

C. Experimental Procedure

Participants were directed to a private counseling room where they engaged in a 10-minute conversation with the avatar about a stressful experience or issue they were currently facing. The conversation was unscripted, and participants were allowed to type freely about their experiences. During the conversation, the avatar responded with appropriate facial expressions and text responses that matched the emotional content of the participant's speech. Participants in the control condition engaged in the same conversation with a non-responsive avatar.

After the conversation, participants were asked to complete a survey that measured their perceived sensory synchronicity, presence, immediacy, interactivity, and similarity with the avatar, as well as their satisfaction with the counseling experience and rapport with the avatar. Participants also rated the degree to which the avatar's facial expressions accurately reflected their emotions during the conversation.

D. Data Analysis

We compared the survey responses between the two groups using independent t-tests and conducted additional analyses to examine the relationship between perceived sensory synchronicity, presence, immediacy, interactivity, and similarity with the avatar, and the outcomes of the counseling experience.

Overall, the data analysis allowed us to determine the effectiveness of our real-time emotion recognition and expression model for avatar-mediated mental health counseling.

IV. Result

Based on our preliminary analysis, it appears that the real-time emotion recognition and expression model proposed for avatar-mediated mental health counseling has a significant positive impact on the emotional experience and rapport building between clients and counselors.

V. Discussion

In this study, we developed a real-time emotion recognition and expression model for avatar-mediated mental health counseling and analyzed its impact on client emotional experience and rapport building. The results showed that the experimental group, which received avatar-mediated counseling with real-time emotional expression, reported a significantly higher sense of presence, immediacy, interactivity, and similarity with the avatar compared to the control group. Additionally, the experimental group reported significantly higher levels of satisfaction, trust, and rapport compared to the control group.

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The results suggest that real-time emotion recognition and expression in avatar-mediated counseling can enhance the sense of presence and immediacy in the counseling session, leading to a higher level of interactivity and rapport building between the client and counselor. Matching the facial expressions of the avatar to the conveyed emotions of the client seems to contribute to the higher level of satisfaction and trust in the counseling session.

However, some limitations should be considered when interpreting the results. First, the sample size of the study was relatively small, and the participants were only college students. Thus, the generalizability of the findings to other populations is limited. Another limitation of this study is its cross-sectional design, which prevents us from making causal inferences about the relationship between avatar-mediated counseling and the emotional experience of the client.

Despite these limitations, this study offers important perspectives into the potential benefits of real-time emotion recognition and expression in avatar-mediated mental health counseling. Future research should aim to replicate and expand upon these findings with larger and more diverse samples and explore the long-term effects of avatar-mediated counseling on client outcomes.

VI. Conclusion

In conclusion, this study aimed to explore the potential of avatar-mediated counseling for enhancing nonverbal communication in mental health counseling. We proposed a real-time emotion recognition and expression model for avatar-mediated counseling and tested its effectiveness on client satisfaction, rapport, and trust. Our results showed that the use of an avatar with emotion recognition and expression capabilities significantly improved client satisfaction and perceived rapport compared to a non-responsive avatar. These findings suggest that avatar-mediated counseling has the potential to enhance the therapeutic relationship and improve the overall effectiveness of mental health counseling. However, further research is needed to investigate the long- term effects of avatar-mediated counseling and to explore its potential in different populations and contexts.

Acknowledgment

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References

[1] S. G. Henry, A. Fuhrel-Forbis, M. A. M. Rogers, and S. Eggly, “Association between nonverbal communication during clinical interactions and outcomes: A systematic review and meta-analysis,” *Patient Education and Counseling*, vol. 86, no. 3, pp. 297–315, 2012.

[2] A. S. Miner, A. Milstein, and J. T. Hancock, “Talking to machines about personal mental health problems,” *JAMA*, vol. 318, no. 13, p. 1217, 2017.

[3] D. L. Roter, R. M. Frankel, J. A. Hall, and D. Sluyter, “The expression of emotion through nonverbal behavior in medical visits. mechanisms and outcomes,” *Journal of General Internal Medicine*, vol. 21, no. S1, 2006.

[4] C. F. Sharpley, A. M. Jeffrey, and T. Mcmah, “Counsellor facial expression and client-perceived rapport,” *Counselling Psychology Quarterly*, vol. 19, no. 4, pp. 343–356, 2006.

[5] A. A. Rizzo, U. Neumann, R. Enciso, D. Fidaleo, and J. Y. Noh, “Performance-driven facial animation: Basic research on human judgments of emotional state in facial avatars,” *CyberPsychology & Behavior*, vol. 4, no. 4, pp. 471–487, 2001.

[6] J. Ku, H. J. Jang, K. U. Kim, J. H. Kim, S. H. Park, J. H. Lee, J. J. Kim, I. Y. Kim, and S. I. Kim, “Experimental results of Affective Valence and arousal to avatar's facial expressions,” *CyberPsychology & Behavior*, vol. 8, no. 5, pp. 493–503, 2005.

[7] M. H., Alkawaz, D., Mohamad, A. H., Basori, and T. Saba, “Blend shape interpolation and FACS for realistic avatar,” *3D Research*, vol.6, pp. 1-10, 2015

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